

NBA Shot Chart Explorer

A Streamlit application for exploring NBA player shooting data and visualizing shot locations on an NBA half court.

The app retrieves historical shot data from the NBA Stats API and allows users to analyze a player's shooting performance by season and season type.

Features

- Select an NBA player
- View individual seasons or career-level shooting data
- Switch between Regular Season and Playoffs
- Interactive NBA shot chart
- Made and missed shot visualization
- Overall field goal percentage
- Three-point percentage
- Shot attempts, makes, and misses
- Field goal percentage by shot distance
- Shooting statistics by court zone
- Raw shot-level data
- Cached API requests for improved performance
- Retry handling for temporary NBA API connection errors

Data Availability

Shot tracking is limited to seasons beginning with the **1996–97 NBA season**.

Career statistics in the application therefore represent available shot data from 1996–97 onward rather than necessarily covering a player's entire NBA career.

Technologies

The application is built with:

- Python
- Streamlit
- pandas
- Matplotlib
- nba_api
- Requests

Installation

Clone or download the project and install the required Python packages:

```
pip install streamlit pandas matplotlib nba_api requests
```

Using a virtual environment is recommended.

For example:

```
python -m venv .venv
```

Activate the environment on macOS/Linux:

```
source .venv/bin/activate
```

Or on Windows:

```
.venv\Scripts\activate
```

Then install the dependencies:

```
pip install streamlit pandas matplotlib nba_api requests
```

Running the Application

From the project directory, run:

```
streamlit run app.py
```

Replace `app.py` with the Python filename containing the application if your file uses a different name.

Streamlit will start a local server and provide an address that can be opened in your browser.

Using the App

1. Select a player from the player dropdown.
2. Select either **Career** or an available season.
3. Choose **Regular Season** or **Playoffs**.

4. Review the player's overall shooting statistics.
5. Explore the shot chart to see where shots were made and missed.
6. Review shooting percentages by distance and court zone.
7. View the underlying shot data at the bottom of the application.

Shot Chart

The shot chart plots each field-goal attempt using coordinates supplied by the NBA Stats API.

- Green circles represent made shots.
- Red X markers represent missed shots.

The application draws an NBA half court over the shot locations to provide geographic context.

Shot Distance Analysis

Shots are grouped by their recorded shot distance.

For each distance, the application displays:

- Makes
- Attempts
- Field goal percentage

Shot Zone Analysis

The application also groups shots into court zones such as:

- Restricted Area
- In The Paint (Non-RA)
- Mid-Range
- Above the Break 3
- Right Corner 3
- Left Corner 3
- Backcourt

The restricted area is recalculated from the shot coordinates using a four-foot radius from the basket rather than relying exclusively on the NBA API's assigned zone.

Data Source

Player information, career statistics, and shot-chart data are retrieved using the `nba_api` Python package, which interfaces with NBA.com statistics endpoints.

Because the application relies on NBA.com's servers, requests may occasionally fail or time out. The application includes retry handling for temporary connection problems.

If data retrieval fails, try using the **Clear cache** button and reload the application.

Disclaimer

This project is intended for educational and basketball analytics purposes.

NBA data is obtained through NBA.com statistics endpoints using the `nba_api` package. Availability and reliability of the data depend on those external services.